

3 (a) Meiosis is one process that contributes to genetic variation.

(i) State **precisely** the stage of meiosis where single chromosomes line up on the equator.

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(ii) Outline the events taking place during anaphase I of meiosis.

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(iii) Describe how crossing over during meiosis leads to genetic variation.

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Mutation also causes genetic variation. Some populations of water hemp, *Amaranthus tuberculatus*, have evolved herbicide resistance as a result of a mutation. This is a problem for farmers as water hemp grows in crop fields, lowering productivity.

Two populations of water hemp were tested for resistance to the herbicide mesotrione. One was a population known to be resistant (control) and the other was a test population, whose resistance was unknown.

- Leaves were removed and immersed in a radioactively labelled solution of mesotrione.
- The leaves absorbed some mesotrione and became radioactive.
- Resistant leaves are able to degrade mesotrione by metabolism.
- The time for 50% of absorbed mesotrione to degrade was calculated by measuring the radioactivity of the leaves.

The results are shown in Table 3.1.

Table 3.1

population of water hemp	mean time for 50% of absorbed mesotrione to degrade / hours	standard deviation
test	27.5	4.75
control	10.1	2.34

- (b) (i) Explain how the results in Table 3.1 show that the two populations differ in their resistance to mesotrione.

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- (ii) Explain why this example of genetic variation is important for natural selection in water hemp populations.

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- (iii) Farmers can send in a sample of leaves of water hemp from their fields to a laboratory to be tested for resistance to mesotrione or other herbicides.

Suggest the benefit of this to a farmer.

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- (c) The null hypothesis states there is no significant difference between the mean times for 50% of absorbed mesotrione to degrade in the two populations.

A t -test can be carried out to compare these two means. The critical value for t at the $p = 0.05$ significance level is 2.23.

- (i) Use the formula in Fig. 3.1 to calculate the value of t .

Show your working.

$$t = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$

Key	
$\bar{x}$	= mean
s	= standard deviation
n_1	= 6 (number of readings for test population)
n_2	= 6 (number of readings for control population)

Fig. 3.1

$t =$ [2]

- (ii) Use your calculated value of t to explain whether the null hypothesis should be accepted or rejected.

accept or reject

explanation

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[Total: 14]